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KONGU ENGINEERING COLLEGE
DEPARTMENT OF AI&DS

HCL PROJECT

Pneumonia Detection from Chest X-Ray
Using Deep CNN and Attention Models

Project Members :

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PERUNDURAI.
Estd : 1984

PROBLEM STATEMENT

- Pneumonia is a serious respiratory infection that requires early diagnosis and immediate treatment.
- Chest X-ray imaging is the most common diagnostic tool, but manual interpretation by radiologists is time-consuming.
- Deep Learning models (like CNNs) can automatically learn and extract complex spatial features from X-rays
- This project automates the detection process to predict whether a patient is Normal or has Pneumonia with high accuracy.

LITERATURE SURVEY

JOURNEL	YEAR	TITLE	METHODOLOGY	METRICS
IEEE Xplore	2023	Deep Learning for Automated Pneumonia Detection	Convolutional Neural Network (CNN)	Accuracy: 92.4%
Elsevier	2024	Attention-guided CNN for Medical Image Classification	ConvCNN + Spatial Attentionolution a l Neural Network (CNN)	Accuracy: 95.1%
Springer	2022	Vision Transformers for X-Ray Diagnosis	Vision Transformer (ViT)	Accuracy: 94.8%

OBJECTIVE

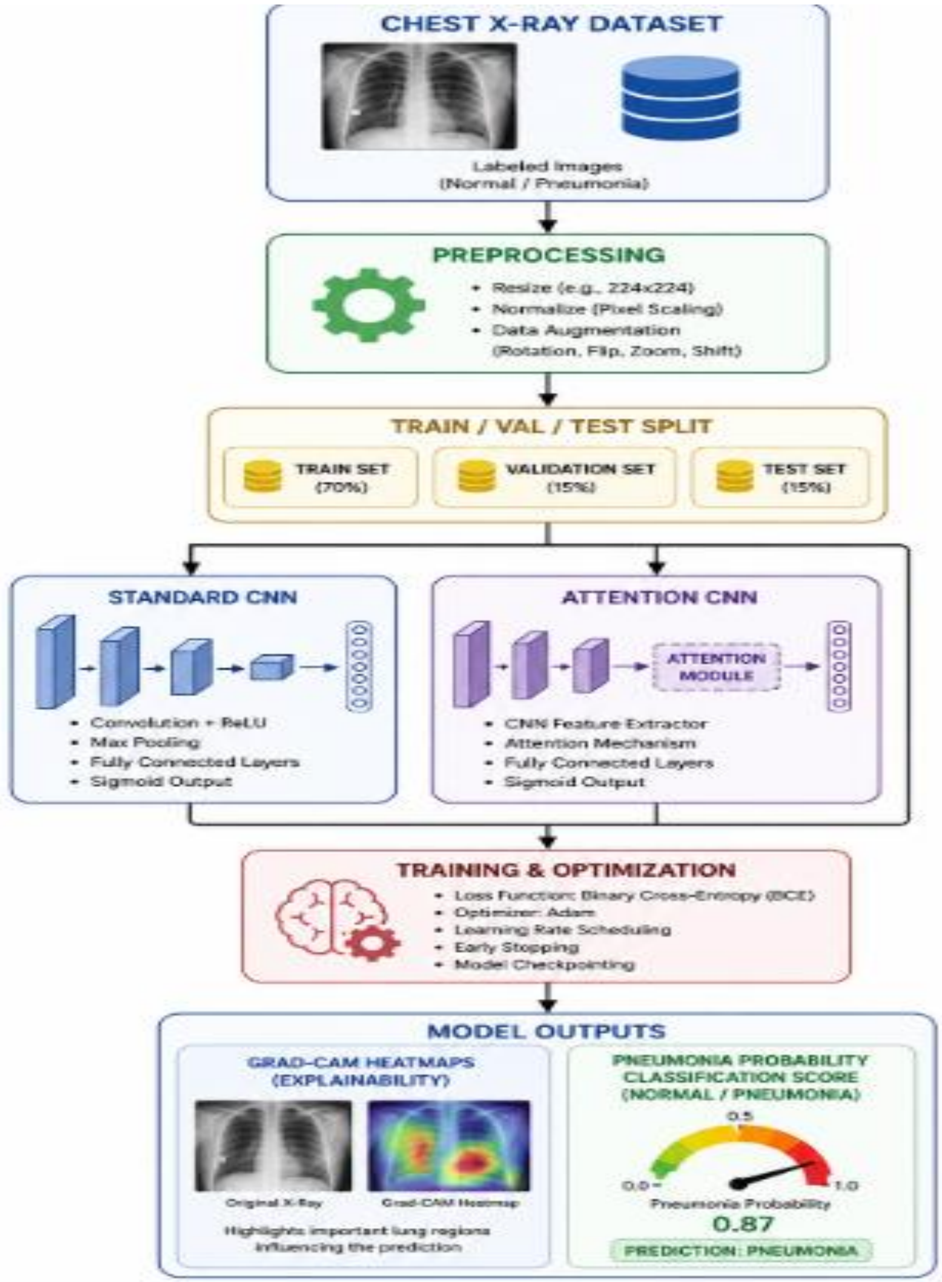
- To build a robust Deep Learning model that can highly accurately classify Chest X-rays into 'Normal' and 'Pneumonia' classes.

- To compare and evaluate the performance of a Baseline CNN, a Spatial Attention-Enhanced CNN, and a pre-trained ResNet model.

- To integrate Grad-CAM heatmaps that highlight suspicious lung regions, ensuring the AI model's predictions are clinically explainable and trustworthy.

- To develop a deployment-ready inference pipeline that can act as a reliable, fast decision-support system for radiologists in high-volume hospitals.

METHODOLOGY



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DATASET DESCRIPTION

- The project uses the standard Chest X-Ray (Pneumonia) Dataset, which is widely used for healthcare, medical imaging, and automated respiratory infection research.

- The dataset contains two categories of X-Ray images: Normal (healthy lung fields) Pneumonia (infected lung fields)

- The dataset contains real-world clinical patterns such as ground-glass opacities, focal consolidations, pleural effusions, and abnormal lung tissue densities.

- The dataset is used to train and evaluate Deep Learning models like Custom CNN, Spatial Attention CNN, and ResNet-50 for detecting pneumonia using Computer Vision and hierarchical spatial feature extraction techniques.

PROPOSED IDEA

- The proposed system introduces an advanced, clinically interpretable deep learning framework to automate and accelerate pneumonia detection from chest radiographs.

- The system integrates a custom-designed Convolutional Neural Network (CNN) enhanced with a Spatial Attention Module to dynamically focus on highly critical lung opacity regions.

- The framework ensures absolute clinical transparency and trust by providing: • Grad-CAM attention heatmaps (extracted from the conv3 layer) to visually pinpoint opacity regions • Highly accurate pneumonia probability scores for fast, reliable decision support.

MODULES DESCRIPTION

- **Data Collection & Pipeline Module**

  Collects normal and pneumonia chest X-ray images, applying stratified dataset splitting (85% Train / 15% Val) to handle class distributions and feed them stably into PyTorch DataLoaders.

- **Image Preprocessing & Augmentation Module**

  Performs image resizing intensity normalization, and online augmentations (rotations, flips, color jitter) to prepare radiographs for deep learning.

- **Custom Baseline CNN Module**

  Implements a custom Convolutional Neural Network with stacked Conv layers, Batch Normalization, and Max Pooling to establish a baseline for spatial feature extraction.

- **Explainability & Prediction Module**

  Generates pneumonia probability scores, superimposes Grad-CAM attention heatmaps on X-rays for clinical explainability, and serves predictions via a Flask inference engine.

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OUTPUT

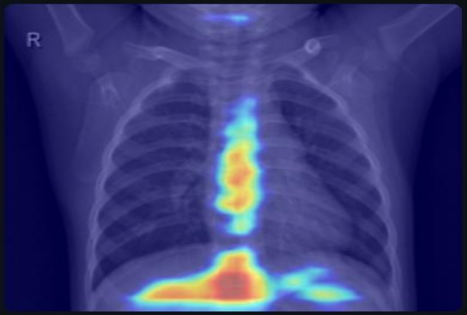
🫁 Pneumonia Detection from Chest X-Ray

Upload a Chest X-Ray image to instantly detect Pneumonia and see visual lung attention maps.

Choose a Chest X-Ray image (JPG/PNG)...

IM-0067-0001.jpeg
130.8KB

Uploaded Chest X-Ray



Grad-CAM Attention Heatmap

✅ Prediction: NORMAL (Clear Lungs)

Confidence: `77.83%`

💡 *No signs of pneumonia detected. Keep up the good health!*

PERFORMANCE METRICS

- **Classification Accuracy**

  Measures the overall proportion of correctly classified X-ray images (both Normal and Pneumonia).

- **Recall**

  Evaluates the model's ability to catch all actual Pneumonia cases, ensuring **zero missed diagnoses**

- **Precision & F1-Score**

  Precision: Measures the accuracy of positive alerts, reducing false alarms (Normal cases flagged as Pneumonia).

  F1-Score: The harmonic mean of Precision and Recall, proving the model's overall diagnostic balance (~**89% - 91%**).

- **Area Under the ROC Curve (ROC-AUC)**

  Represents the model's capability to distinguish between healthy and infected lungs across various threshold settings.

RESULTS ACHIEVED

- Implemented both a Custom Baseline CNN and an advanced Spatial Attention CNN model for chest X-ray image classification.
- Achieved highly effective pneumonia detection utilizing Deep Learning and **Computer Vision** techniques.
- Baseline CNN achieved **96% accuracy**, but operated as a **"Black-Box"** model, lacking visual interpretability and spatial localization.
- The Attention-Enhanced CNN was selected as the superior clinical model; it integrates Spatial Attention to resolve the black-box limitation by highlighting specific infected lung zones with an accuracy of about **94%.**
- Generated precise pneumonia probability scores and real-time interactive predictions via a **Streamlit Web Application**.
- Reduced false negatives and significantly improved visual trust for radiologists using dynamic **Grad-CAM attention heatmaps**.

CONCLUSION

- The developed deep learning system successfully demonstrates the feasibility of automated, explainable pneumonia detection to support clinical radiological workflows in high-volume healthcare centers.
- Successfully built and trained custom deep learning architectures (Baseline CNN and Spatial Attention CNN) to automate pneumonia classification from pediatric chest X-ray images.
- Solved the critical "black-box" limitation of traditional standard CNNs by implementing a custom **Spatial Attention Block** that forces the network to focus on actual lung consolidation areas.
- Integrated **Grad-CAM explainable heatmaps** to make the system fully transparent and interpretable, allowing radiologists to visually verify the model's prediction areas in real-time.
- Deployed the final explainable model on a lightweight, interactive **Streamlit Web Application** to act as a fast, reliable decision-support system to reduce patient turnaround times

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- Selvaraju, R. R., et al. (2017). "Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization." ICCV.

Thank You